

PROJECT TURNCOAT

MAPPING THE ADVERSARY'S HYPERSONIC INDUSTRIAL BASE

SYSTEM OBJECTIVE // CODE: TURNCOAT

THE PROBLEM & CHALLENGE

Adversary hypersonic supply chains are heavily obfuscated across fragmented sources. We must deploy NLP pipelines to ingest arXiv scientific research, Google Patents data, and international trade records to map and expose critical structural vulnerabilities.

- **ACADEMIC OSINT & ARXIV API**
NLP PARSING OF FOREIGN PHYSICS & AEROSPACE PAPERS
- **PATENT DATA EXTRACTION**
RESOLVING PROPRIETARY SCRAMJET & METALLURGY DESIGNS
- **TRADE & SUPPLY CHAIN MAPPING**
CORRELATING GLOBAL LOGISTICS TO PINPOINT CHOKEPOINTS

HIGH-LEVEL SOFTWARE BILL OF MATERIALS & CORE DEPENDENCIES

[CORE APPLICATION]

Nim-based web application designed for high-performance open-source intelligence analysis and network topology mapping.

Nim 2.2.x Web App

LICENSE: MIT

Dependency Control

23 Nim packages pinned to exact versions, Git revisions, and checksums.

[ENVIRONMENT & SERVICES]

Deployment Environment

SQLite 3.38+, OpenSSL, and optional Poppler pdftotext utilities.

External Service Integrations

arXiv Preprint Metadata Integration

Google Patents Patent Data API

OpenAI-Compatible Inference Endpoint

CORE NIM PACKAGES // DEPS

HappyX 4.7.4

Web Server & User Interface Architecture

Grim 0.3.1

Adversary Graph & Network Processing

db_connector 0.1.0

SQLite High-Fidelity Database Integration

NimYAML 0.16.0

Serialized Graph Configuration
Compatibility

htmlparser 0.1.0

Automated HTML Document Parsing

checksums 0.2.2

Content Cryptographic Hashing

BUNDLED BROWSER ASSETS

D3 7.9.0

Interactive Graph Rendering

LICENSED VIA ISC

markdown-it 15.0.1

Structured Markdown Parser

LICENSED VIA MIT

IBM Plex Fonts

Tactical Monospaced Interface

LICENSED VIA OFL-1.1

[PREDICTIVE ANALYSIS]

Advanced forecasting models identify and project target hardware platforms, critical software stacks, and development methodologies.

AUTOMATED NLP PIPELINES EXTRACT ACTIONABLE INTELLIGENCE FROM FRAGMENTED DATA

[PHASE 01 // FOREIGN SCIENTIFIC RESEARCH]

Natural Language Processing algorithms systematically parse simulated physics and aerospace engineering papers from international repositories to identify emerging hypersonic research trends.

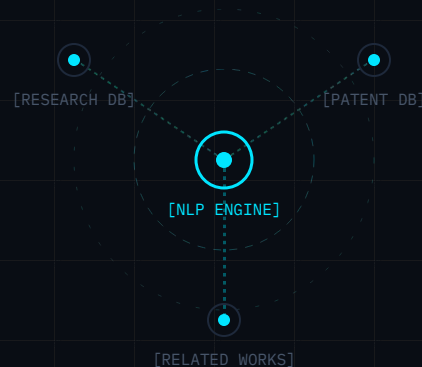
[PHASE 02 // GLOBAL PATENT ANALYSIS]

Automated ingestion of international patent databases reveals proprietary advancements in high-stress metallurgy and scramjet propulsion systems.

[PHASE 03 // CO-CITATION NETWORKS]

Citation and co-authorship graph analysis maps academic researchers directly to their foundational discoveries and related publications, tracing the lineage of scientific innovation.

INTEL KNOWLEDGE GRAPH // RESOLVED



Knowledge Graphs Transform Unstructured Data into High-Fidelity Network Maps

[ARXIV & PATENT RESOLUTION]

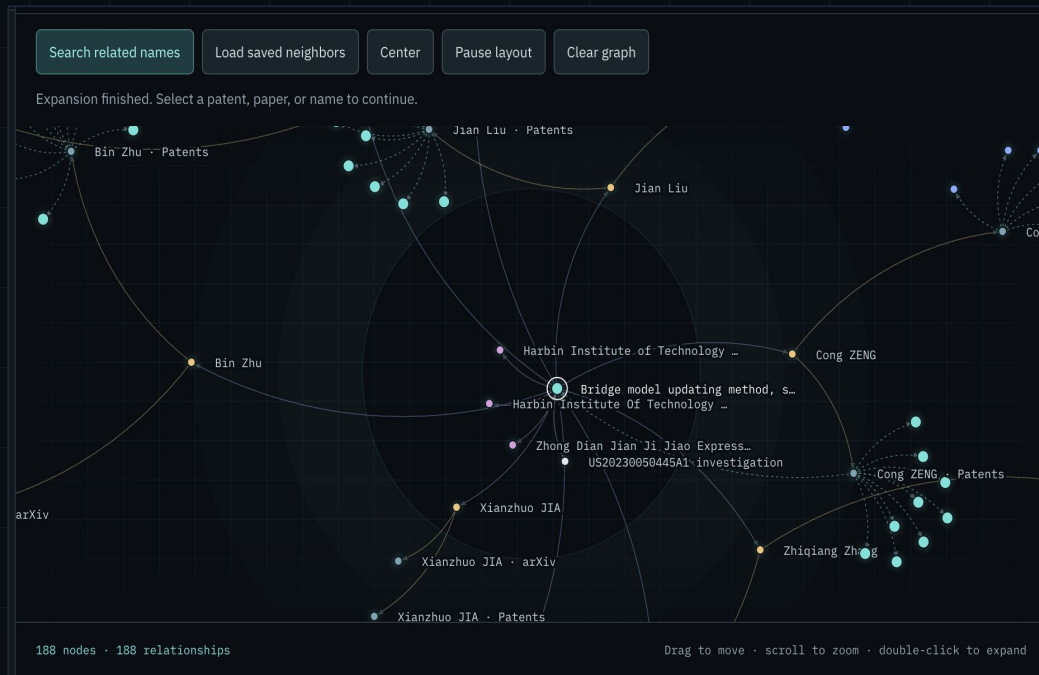
The system automatically reconciles and extracts fragmented author and inventor identities directly from arXiv preprints and Google Patents database records to build verified technical profiles.

[ARXIV & PATENT ANALYSIS]

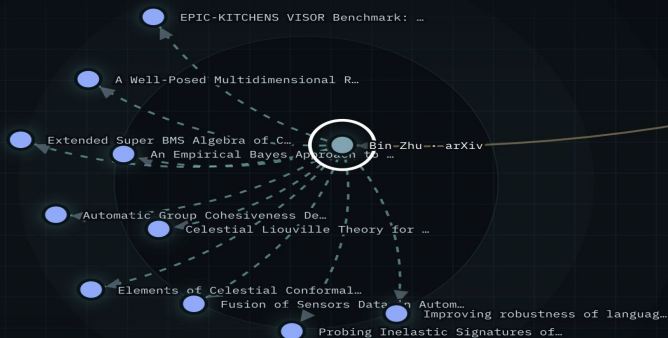
Unstructured academic preprints and patent publications are converted into an explorable network, mapping citation lines and shared disclosures between researchers and corporate entities.

[CROSS-CITATION CORRELATION]

Graph algorithms track co-citations and joint developments across arXiv preprints and Google Patents filings, identifying foundational intellectual IP chokepoints.



PRECISION INTELLIGENCE ENABLES TARGETED INTERVENTIONS TO DISRUPT HYPERSONIC DEVELOPMENT



[CRITICAL CHOKEPOINT IDENTIFICATION]

STRATEGIC CHOKEPOINTS

The intelligence graph provides INDOPACOM and Strategic Command with exact vulnerabilities within the adversary's scaling pipeline.

[01] ECONOMIC SANCTIONS

[STATUS: ACTIVE]

Identifies specific front companies and financial networks for precision economic targeting and treasury sanctions.

[02] EXPORT CONTROLS

[RISK: SEVERE]

Informs the restriction of bespoke manufacturing equipment and specialized raw materials, degrading the adversary's production capacity.

[03] INFLUENCE OPERATIONS

[VECTOR: DISRUPT]

Maps key personnel and research institutions to support targeted disruption and counter-intelligence operations.

MISSION-CONTROL INTERFACE ACCELERATES COMPLEX INTELLIGENCE ANALYSIS

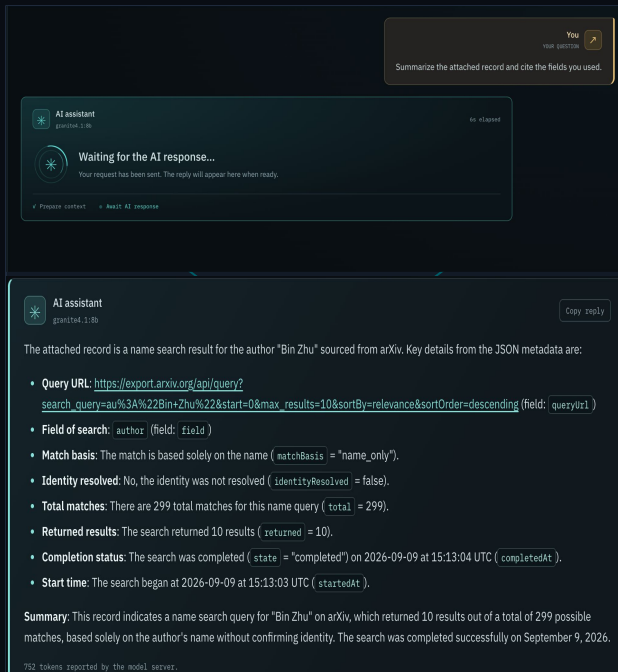
+ [QUERY: LOCAL AI PIPELINE / FOREIGN TRANSLATION / KEYWORDS]

[LOCAL AI PLATFORM]

A fully offline, air-gapped language model architecture designed to process sensitive intelligence locally without external API calls, maintaining complete operational security.

[FOREIGN TRANSLATION]

Automated neural translation engines convert raw foreign aerospace preprints, patents, and military research papers into English while preserving complex technical nomenclature.



AI COGNITIVE EXTRACTION PANEL

[PDF SUMMARIZATION]

Generates multi-level, structured summaries of complex technical documents, extracting core scientific methodologies and testing parameters instantly.

[KEYWORD ANALYSIS]

Sifts through translated corpora to isolate strategic keywords, military-industrial codes, and institutional names to map underlying development networks.

DOWN-SELECT ROADMAP: ADVANCED COGNITIVE ANALYTICS & SUPPLY CHAIN PREDICTION

[PHASE I: ANALYTICS, TAGGING & COGNITIVE INFERENCE]

Improved Analytics & Tagging

Integration of automated NLP taxonomy generators to auto-tag raw research with military codes and standardized aerospace nomenclature.

Enhanced AI Inference

Transitioning models to a multi-agent routing pipeline, dramatically increasing local extraction speed and technical keyword classification accuracy.

[PHASE II: END-TO-END SUPPLY CHAIN PREDICTION]

Full Supply Chain Mapping

Scaling node-link graph capabilities from isolated researchers to map the entire industrial base—correlating precursors, tooling, and end-stage delivery.

Chokepoint Discovery

Identifying critical structural dependencies where single-source manufacturers or specialized alloys threaten adversary scaling efforts.

COUNTERMEASURE RECOMMENDATION ENGINE

[FUTURE TARGETS]

[01] TECHNOLOGICAL DEGRADATION

Recommends precision exports control protocols targeting advanced carbon-carbon composites and bespoke sintering furnaces identified in patent data.

[02] ECONOMIC INTERVENTION

Tracks funding paths of front companies using localized co-citation anomalies to recommend structured banking blockades and financial penalties.

[03] OPEN DATA DISSEMINATION

Provides structured, classified graph exports to unified combatant commands, enabling external agencies to model independent kinetic/cyber options.